

Frank — AMC Drill M8 · route inspection

25 Jun 2026 | ANSWER KEY & method — for the parent

1. Answer: **120 km** [M8 · twin of real 2018 Q24 · 5 marks]

Working: all 4 corners have 3 fences (odd) → re-walk the two shortest sides. Fence total = $2 \times 16 + 2 \times 12 + 2 \times 20 = 96$ km; minimum extra = $2 \times 12 = 24$ km; shortest route = $96 + 24 = 120$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

2. Answer: **140 km** [M8 · twin of real 2018 Q24 · 5 marks]

Working: all 4 corners have 3 fences (odd) → re-walk the two shortest sides. Fence total = $2 \times 24 + 2 \times 10 + 2 \times 26 = 120$ km; minimum extra = $2 \times 10 = 20$ km; shortest route = $120 + 20 = 140$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

3. Answer: **180 km** [M8 · twin of real 2018 Q24 · 5 marks]

Working: all 4 corners have 3 fences (odd) → re-walk the two shortest sides. Fence total = $2 \times 18 + 2 \times 24 + 2 \times 30 = 144$ km; minimum extra = $2 \times 18 = 36$ km; shortest route = $144 + 36 = 180$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

4. Answer: **192 km** [M8 · twin of real 2018 Q24 · 5 marks]

Working: all 4 corners have 3 fences (odd) → re-walk the two shortest sides. Fence total = $2 \times 30 + 2 \times 16 + 2 \times 34 = 160$ km; minimum extra = $2 \times 16 = 32$ km; shortest route = $160 + 32 = 192$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.

5. Answer: **300 km** [M8 · twin of real 2018 Q24 · 5 marks]

Working: all 4 corners have 3 fences (odd) → re-walk the two shortest sides. Fence total = $2 \times 40 + 2 \times 30 + 2 \times 50 = 240$ km; minimum extra = $2 \times 30 = 60$ km; shortest route = $240 + 60 = 300$ km.

★ Count how many fences meet at each junction: if any has an ODD number you cannot do it without repeats. Pair up the odd junctions and re-walk only the cheapest connecting fences; answer = total fence length + that minimum extra.

Trap: just adding up all the fences and forgetting you must double back on the odd junctions.